

1. CEM

Задача:

$$x^* \in \arg \max_x S(x), \quad x \sim q_\theta(x).$$

Редкое событие / элита:

$$p^*(x) = p(x \mid S(x) \geq \gamma) = \frac{p(x) \mathbf{1}\{S(x) \geq \gamma\}}{\mathbb{P}(S(X) \geq \gamma)}.$$

Проекция на семейство:

$$\theta^* = \arg \min_{\theta} \text{KL}(p^* \| q_\theta) = \arg \max_{\theta} \mathbb{E}_{p^*} \log q_\theta(X).$$

Практическое обновление:

$$\mathcal{E}_k = \{x_i : S(x_i) \geq \gamma_k\}, \quad \theta_{k+1} = \arg \max_{\theta} \sum_{x_i \in \mathcal{E}_k} \log q_\theta(x_i).$$

Для $q_\theta = \mathcal{N}(\mu, \Sigma)$:

$$\mu^+ = \frac{1}{|\mathcal{E}|} \sum_{x_i \in \mathcal{E}} x_i, \quad \Sigma^+ = \frac{1}{|\mathcal{E}|} \sum_{x_i \in \mathcal{E}} (x_i - \mu^+)(x_i - \mu^+)^{\top}.$$

Сглаживание:

$$\theta_{k+1} \leftarrow \alpha \theta^+ + (1 - \alpha) \theta_k.$$

RL-вариант:

$$\tau \sim q_\theta(\tau), \quad S(\tau) = R(\tau) = \sum_t \gamma^t r_t, \quad \theta^+ = \arg \max_{\theta} \sum_{\tau_i \in \mathcal{E}} \log q_\theta(\tau_i).$$

2. Беллман, PI/VI

MDP:

$$M = (\mathcal{S}, \mathcal{A}, p, r, \gamma), \quad p(s' \mid s, a), \quad \pi(a \mid s).$$

Ядро политики:

$$P_\pi(s' \mid s) = \sum_a \pi(a \mid s) p(s' \mid s, a).$$

Функции ценности:

$$V^\pi(s) = \mathbb{E}_\pi \sum_{t \geq 0} \gamma^t r_t \mid s_0 = s, \quad Q^\pi(s, a) = \mathbb{E}_\pi \sum_{t \geq 0} \gamma^t r_t \mid s_0 = s, a_0 = a.$$

Связь V, Q :

$$V^\pi(s) = \sum_a \pi(a \mid s) Q^\pi(s, a), \\ Q^\pi(s, a) = r(s, a) + \gamma \sum_{s'} p(s' \mid s, a) V^\pi(s').$$

Беллман для политики:

$$V^\pi(s) = \sum_a \pi(a \mid s) \left[r(s, a) + \gamma \sum_{s'} p(s' \mid s, a) V^\pi(s') \right].$$

В матричном виде:

$$V^\pi = r_\pi + \gamma P_\pi V^\pi, \quad V^\pi = (I - \gamma P_\pi)^{-1} r_\pi.$$

Оптимальность:

$$V^*(s) = \max_a \left[r(s, a) + \gamma \sum_{s'} p(s' \mid s, a) V^*(s') \right], \\ Q^*(s, a) = r(s, a) + \gamma \sum_{s'} p(s' \mid s, a) \max_{a'} Q^*(s', a').$$

Жадная политика:

$$\pi^*(s) \in \arg \max_a Q^*(s, a).$$

Оператор Беллмана - сжатие:

$$\|BV - BU\|_\infty \leq \gamma \|V - U\|_\infty.$$

Policy Evaluation:

$$V_{k+1} = r_\pi + \gamma P_\pi V_k.$$

Policy Improvement:

$$\pi_{k+1}(s) \in \arg \max_a \left[r(s, a) + \gamma \sum_{s'} p(s' \mid s, a) V^{\pi_k}(s') \right].$$

Value Iteration:

$$V_{k+1}(s) = \max_a \left[r(s, a) + \gamma \sum_{s'} p(s' \mid s, a) V_k(s') \right].$$

3. MC, TD, Q-learning

Return:

$$G_t = \sum_{k=t}^{T-1} \gamma^{k-t} r_k.$$

Инкрементальное среднее:

$$\hat{Q}_{n+1} = \hat{Q}_n + \alpha_n (G - \hat{Q}_n), \quad \sum_n \alpha_n = \infty, \quad \sum_n \alpha_n^2 < \infty.$$

MC-оценивание:

$$Q(s, a) \leftarrow Q(s, a) + \alpha (G_t - Q(s, a)).$$

TD(0) для V :

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t), \quad V(s_t) \leftarrow V(s_t) + \alpha \delta_t.$$

SARSA:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha [r_t + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)].$$

Q-learning:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha [r_t + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t)].$$

Expected SARSA:

$$y_t = r_t + \gamma \sum_{a'} \pi(a' \mid s_{t+1}) Q(s_{t+1}, a').$$

ε -greedy:

$$\pi(a \mid s) = \begin{cases} 1 - \varepsilon + \varepsilon / |\mathcal{A}|, & a \in \arg \max_b Q(s, b), \\ \varepsilon / |\mathcal{A}|, & \text{иначе.} \end{cases}$$

MC: без смещения, большая дисперсия. TD: смещённая оценка, малая дисперсия.

4. DQN и модификации

DQN-target:

$$y = r + \gamma(1 - d) \max_{a'} Q_{\theta^-}(s', a').$$

Лосс:

$$L(\theta) = \mathbb{E}_{(s, a, r, s', d) \sim \mathcal{D}} (Q_\theta(s, a) - y)^2.$$

Replay buffer:

$$(s_t, a_t, r_t, s_{t+1}, d_t) \rightarrow \mathcal{D}, \quad B \sim \text{Unif}(\mathcal{D}).$$

Target network:

$$\theta^- \leftarrow \theta \quad \text{каждые } K \text{ шагов,} \quad \text{или} \quad \theta^- \leftarrow \tau \theta + (1 - \tau) \theta^-.$$

Double DQN:

$$a^* = \arg \max_{a'} Q_\theta(s', a'), \quad y = r + \gamma(1 - d) Q_{\theta^-}(s', a^*).$$

PER:

$$\delta_i = y_i - Q(s_i, a_i), \quad p_i \propto (|\delta_i| + \varepsilon)^\alpha, \quad P(i) = \frac{p_i}{\sum_j p_j},$$

$$w_i = \left(\frac{1}{NP(i)} \right)^\beta, \quad L = \mathbb{E}_i w_i \delta_i^2.$$

Dueling DQN:

$$Q(s, a) = V(s) + A(s, a) - \frac{1}{|\mathcal{A}|} \sum_b A(s, b).$$

NoisyNet:

$$W = \mu_W + \sigma_W \odot \varepsilon_W.$$

Multi-step DQN:

$$y_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k r_{t+k} + \gamma^n (1 - d_{t+n}) \max_{a'} Q_{\theta^-}(s_{t+n}, a).$$

DRQN / memory:

$$h_t = f_\theta(o_t, h_{t-1}), \quad Q_t = Q_\theta(h_t, a).$$

5. Distributional RL, QR-DQN, IQN

Случайная доходность:

$$Z^\pi(s, a) = \sum_{t \geq 0} \gamma^t r_t, \quad Q^\pi(s, a) = \mathbb{E} Z^\pi(s, a).$$

Distributional Bellman:

$$Z^\pi(s, a) \stackrel{D}{=} r(s, a) + \gamma Z^\pi(s', a'), \quad s' \sim p(\cdot \mid s, a), \quad a' \sim \pi(\cdot \mid s').$$

Оптимальный вариант:

$$Z^*(s, a) \stackrel{D}{=} r + \gamma Z^*(s', a^*), \quad a^* = \arg \max_b \mathbb{E} Z^*(s', b).$$

Wasserstein:

$$W_p(U, V) = \left(\inf_{\mu \in \Gamma(U, V)} \mathbb{E}_{(x, y) \sim \mu} |x - y|^p \right)^{1/p}.$$

QR-DQN аппроксимация:

$$Z_\theta(s, a) = \frac{1}{N} \sum_{i=1}^N \delta_{\theta_i}(s, a), \quad \tau_i = \frac{i - 1/2}{N}.$$

Таргет-атомы:

$$y_j = r + \gamma \theta_j^-(s', a^*), \quad a^* = \arg \max_a \frac{1}{N} \sum_j \theta_j(s', a).$$

Квантильный Huber:

$$\rho_\tau^\kappa(u) = |\tau - \mathbf{1}\{u < 0\}| L_\kappa(u), \\ L_\kappa(u) = \begin{cases} \frac{1}{2} u^2, & |u| \leq \kappa, \\ \kappa(|u| - \frac{1}{2} \kappa), & |u| > \kappa. \end{cases}$$

Лосс QR:

$$L = \frac{1}{N^2} \sum_{i,j} \rho_{\tau_i}^\kappa(y_j - \theta_i(s, a)).$$

IQN:

$$\tau \sim U[0, 1], \quad Z_\theta(s, a; \tau) = F_{Z(s, a)}^{-1}(\tau),$$

$$Q(s, a) \approx \frac{1}{K} \sum_{k=1}^K Z_\theta(s, a; \tau_k), \quad \tau_k \sim U[0, 1].$$

6. RND, ICM

Общая внутренняя награда:

$$r_t = r_t^{\text{extr}} + \beta r_t^{\text{intr}}.$$

Разложение для фиксированной политики:

$$Q^\pi = Q_{\text{extr}}^\pi + \beta Q_{\text{intr}}^\pi, \quad V^\pi = V_{\text{extr}}^\pi + \beta V_{\text{intr}}^\pi.$$

Count-based:

$$r^{\text{intr}}(s) = \frac{1}{\sqrt{N(s)}} \quad \text{или} \quad \frac{1}{N(s)}.$$

RND:

$$f : \mathcal{S} \rightarrow \mathbb{R}^d \text{ fixed}, \quad \hat{f}_\theta : \mathcal{S} \rightarrow \mathbb{R}^d \text{ trained},$$

$$L_{\text{RND}}(\theta) = \|\hat{f}_\theta(s) - f(s)\|_2^2, \quad r^{\text{intr}}(s) = \|\hat{f}_\theta(s) - f(s)\|_2^2.$$

ICM encoder:

$$\phi_t = \phi(s_t), \quad \phi_{t+1} = \phi(s_{t+1}).$$

Inverse dynamics:

$$\hat{a}_t = g_\eta(\phi_t, \phi_{t+1}), \quad L_I = \ell(\hat{a}_t, a_t).$$

Forward dynamics:

$$\hat{\phi}_{t+1} = F_\psi(\phi_t, a_t), \quad L_F = \frac{1}{2} \|\hat{\phi}_{t+1} - \phi_{t+1}\|_2^2.$$

ICM loss / reward:

$$L_{\text{ICM}} = (1 - \lambda) L_I + \lambda L_F, \quad r_t^{\text{intr}} = \eta L_F.$$

7. Policy Gradient, REINFORCE

Objective:

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \sum_{t \geq 0} \gamma^t r_t.$$

Log-derivative trick:

$$\nabla_\theta p_\theta(x) = p_\theta(x) \nabla_\theta \log p_\theta(x).$$

Policy Gradient theorem:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \sum_{t \geq 0} \gamma^t \nabla_\theta \log \pi_\theta(a_t | s_t) Q^\pi_\theta(s_t, a_t).$$

Через d^π :

$$\nabla_\theta J = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d^\pi, a \sim \pi} \nabla_\theta \log \pi_\theta(a | s) Q^\pi(s, a).$$

Discounted visitation:

$$d^\pi(s) = (1 - \gamma) \sum_{t \geq 0} \gamma^t \mathbb{P}(s_t = s | \pi).$$

Baseline:

$$\begin{aligned} \mathbb{E}_{a \sim \pi} \nabla_\theta \log \pi_\theta(a | s) b(s) &= 0, \\ \nabla J &= \frac{1}{1 - \gamma} \mathbb{E} \nabla \log \pi_\theta(a | s) (Q^\pi(s, a) - b(s)). \end{aligned}$$

Advantage:

$$A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s).$$

REINFORCE:

$$G_t = \sum_{k=t}^{T-1} \gamma^{k-t} r_k, \quad \hat{g} = \sum_t \gamma^t \nabla_\theta \log \pi_\theta(a_t | s_t) G_t.$$

With baseline:

$$\hat{g} = \sum_t \gamma^t \nabla_\theta \log \pi_\theta(a_t | s_t) (G_t - b(s_t)).$$

Entropy regularization:

$$J_H = J + \beta \mathbb{E}_s H(\pi(\cdot | s)), \quad H(\pi) = -\mathbb{E}_\pi \log \pi.$$

8. Actor-Critic, A2C, GAE

Critic TD-error:

$$\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t).$$

Actor update:

$$\nabla_\theta J \approx \mathbb{E} \nabla_\theta \log \pi_\theta(a_t | s_t) \hat{A}_t.$$

Critic loss:

$$L_V(\phi) = \mathbb{E} \left(V_\phi(s_t) - \hat{V}_t \right)^2.$$

One-step AC:

$$\hat{A}_t = \delta_t, \quad \hat{V}_t = r_t + \gamma V_\phi(s_{t+1}).$$

n -step return:

$$G_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k r_{t+k} + \gamma^n V(s_{t+n}).$$

n -step advantage:

$$\hat{A}_t^{(n)} = G_t^{(n)} - V(s_t).$$

A2C total loss:

$$L(\theta, \phi) = -\mathbb{E} \log \pi_\theta(a_t | s_t) \hat{A}_t + c_v \mathbb{E} (V_\phi(s_t) - \hat{V}_t)^2 - c_H \mathbb{E} H(\pi_\theta(\cdot | s_t)).$$

GAE:

$$\hat{A}_t^{\text{GAE}(\gamma, \lambda)} = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}.$$

Equivalent mixture:

$$\hat{A}_t^{\text{GAE}} = (1 - \lambda) \sum_{n=1}^{\infty} \lambda^{n-1} \hat{A}_t^{(n)}.$$

Target for critic:

$$\hat{V}_t = \hat{A}_t + V_\phi(s_t).$$

9. TRPO

Relative performance identity:

$$J(\pi) - J(\pi_{old}) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d^\pi, a \sim \pi} A^{\pi_{old}}(s, a).$$

Surrogate:

$$L_{\pi_{old}}(\pi) = J(\pi_{old}) + \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d^{\pi_{old}}, a \sim \pi} A^{\pi_{old}}(s, a).$$

Importance sampling:

$$L(\theta) = \mathbb{E}_{s, a \sim \pi_{old}} \frac{\pi_\theta(a | s)}{\pi_{old}(a | s)} A^{\pi_{old}}(s, a).$$

TRPO constrained problem:

$$\max_\theta L(\theta) \quad \text{s.t.} \quad \mathbb{E}_{s \sim d^{\pi_{old}}} \text{KL}(\pi_{old}(\cdot | s) \| \pi_\theta(\cdot | s)) \leq \delta.$$

Lower bound form:

$$J(\pi) \geq L_{\pi_{old}}(\pi) - C D_{\text{TV}}^{\max}(\pi_{old}, \pi)^2.$$

Quadratic approximation:

$$L(\theta) \approx g^\top (\theta - \theta_{old}), \quad \bar{D}_{\text{KL}} \approx \frac{1}{2} (\theta - \theta_{old})^\top H (\theta - \theta_{old}).$$

Natural gradient step:

$$\Delta \theta = \sqrt{\frac{2\delta}{g^\top H^{-1} g}} H^{-1} g.$$

10. Bias-variance, GAE, PPO

MC / n -step / TD:

$$\hat{A}_t^{MC} = \sum_{k=0}^{\infty} \gamma^k r_{t+k} - V(s_t),$$

$$\hat{A}_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k r_{t+k} + \gamma^n V(s_{t+n}) - V(s_t),$$

$$\hat{A}_t^{(1)} = r_t + \gamma V(s_{t+1}) - V(s_t) = \delta_t.$$

Bias-variance: MC $\downarrow$ bias, $\uparrow$ variance; TD $\uparrow$ bias, $\downarrow$ variance. GAE:

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t), \quad \hat{A}_t = \sum_{l \geq 0} (\gamma \lambda)^l \delta_{t+l}.$$

PPO ratio:

$$\rho_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{old}(a_t | s_t)}.$$

PPO-Clip:

$$L^{CLIP}(\theta) = \mathbb{E}_t \min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_t \right).$$

Full PPO loss:

$$L = L^{CLIP} - c_v L_V + c_H H, \quad L_V = \mathbb{E}_t (V_\phi(s_t) - \hat{V}_t)^2.$$

Clipped value loss:

$$V_\phi^{clip} = V_{old} + \text{clip}(V_\phi - V_{old}, -\varepsilon_v, \varepsilon_v),$$

$$L_V = \mathbb{E}_t \max \left((V_\phi - \hat{V})^2, (V_\phi^{clip} - \hat{V})^2 \right).$$

KL-penalty PPO:

$$\max_\theta \mathbb{E}_t \rho_t \hat{A}_t - \beta \mathbb{E}_t \text{KL}(\pi_{old} \| \pi_\theta).$$

11. DDPG, TD3

Deterministic Policy Gradient:

$$\nabla_\theta J(\pi_\theta) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d^\pi} \nabla_\theta \left[\nabla_\theta \pi_\theta(s)^\top \nabla_a Q^\pi(s, a) |_{a=\pi_\theta(s)} \right].$$

Critic target DDPG:

$$y = r + \gamma(1 - d) Q_\omega - (s', \pi_{\theta-}(s')).$$

Critic loss:

$$L_Q(\omega) = \mathbb{E} (Q_\omega(s, a) - y)^2.$$

Actor objective:

$$J_\pi(\theta) = \mathbb{E}_{s \sim \mathcal{D}} Q_\omega(s, \pi_\theta(s)), \quad \theta \leftarrow \theta + \alpha \nabla_\theta J_\pi.$$

Exploration:

$$a_t = \pi_\theta(s_t) + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2 I) \quad \text{или} \quad \text{OU-noise.}$$

Soft target update:

$$\omega^- \leftarrow \tau \omega + (1 - \tau) \omega^-, \quad \theta^- \leftarrow \tau \theta + (1 - \tau) \theta^-.$$

TD3 target policy smoothing:

$$\tilde{a}' = \pi_{\theta-}(s') + \text{clip}(\varepsilon, -c, c), \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I).$$

Clipped double Q:

$$y = r + \gamma(1 - d) \min_{i=1,2} Q_{\omega_i^-}(s', \tilde{a}').$$

Delayed policy updates:

θ обновляется раз в d шагов критика.

12. Entropy RL, SAC

Entropy:

$$H(\pi(\cdot | s)) = -\mathbb{E}_{a \sim \pi} \log \pi(a | s).$$

Maximum Entropy RL:

$$J_{soft}(\pi) = \mathbb{E}_\pi \sum_{t \geq 0} \gamma^t [r_t + \alpha H(\pi(\cdot | s_t))] = \mathbb{E}_\pi \sum_t \gamma^t [r_t - \alpha \log \pi(a_t | s_t)].$$

Soft Q, V :

$$Q_{soft}^\pi(s, a) = r(s, a) + \gamma \mathbb{E}_{s'} V_{soft}^\pi(s'),$$

$$V_{soft}^\pi(s) = \mathbb{E}_{a \sim \pi} \left[Q_{soft}^\pi(s, a) - \alpha \log \pi(a | s) \right].$$

Soft Bellman:

$$Q_{soft}^\pi(s, a) = r + \gamma \mathbb{E}_{s', a'} \left[Q_{soft}^\pi(s', a') - \alpha \log \pi(a' | s') \right].$$

Soft policy improvement:

$$\max_\pi \mathbb{E}_{a \sim \pi} [Q(s, a) - \alpha \log \pi(a | s)] \quad \Leftrightarrow \quad \min_\pi \text{KL} \left(\pi(\cdot | s) \left\| \frac{\exp(Q(s, \cdot)/\alpha)}{Z(s)} \right. \right).$$

SAC critic target:

$$y = r + \gamma(1 - d) \left[\min_i Q_{\omega_i^-}(s', a') - \alpha \log \pi_\theta(a' | s') \right], \quad a' \sim \pi_\theta(\cdot | s').$$

Critic loss:

$$L_Q = \sum_i \mathbb{E} (Q_{\omega_i}(s, a) - y)^2.$$

Actor loss:

$$L_\pi(\theta) = \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta} \left[\alpha \log \pi_\theta(a | s) - \min_i Q_{\omega_i}(s, a) \right].$$

Temperature tuning:

$$L(\alpha) = \mathbb{E}_{a \sim \pi_\theta} [-\alpha (\log \pi_\theta(a | s) + \bar{H})].$$

Reparameterization:

$$a = f_\theta(\epsilon; s), \quad \epsilon \sim \mathcal{N}(0, I).$$

13. Imitation, offline RL, GCL, GAIL

Behavior cloning:

$$\theta^* = \arg \max_{\theta} \sum_{(s,a) \in \mathcal{D}_E} \log \pi_{\theta}(a \mid s).$$

Offline RL dataset:

$$\mathcal{D} = \{(s, a, r, s')\} \sim \mu(a \mid s), \quad \text{без нового online-сбора.}$$

IRL:

$$\mathcal{D}_E = \{\tau_i^E\}, \quad r_{\theta}(s, a) \text{ восстановить.}$$

MaxEnt trajectory model:

$$p_{\theta}(\tau) = \frac{1}{Z(\theta)} \exp(R_{\theta}(\tau)) \prod_t p(s_{t+1} \mid s_t, a_t), \quad R_{\theta}(\tau) = \sum_t r_{\theta}(s_t, a_t).$$

GCL log-likelihood:

$$\log p_{\theta}(\tau) = R_{\theta}(\tau) - \log Z(\theta) + \text{const.}$$

Gradient:

$$\nabla_{\theta} \ell = \mathbb{E}_{\tau \sim \mathcal{D}_E} \nabla_{\theta} R_{\theta}(\tau) - \mathbb{E}_{\tau \sim p_{\theta}} \nabla_{\theta} R_{\theta}(\tau).$$

Occupancy measure:

$$\rho_{\pi}(s, a) = \pi(a \mid s) \sum_{t \geq 0} \gamma^t \mathbb{P}(s_t = s \mid \pi).$$

GAIL objective:

$$\min_{\pi} \max_D \mathbb{E}_{(s,a) \sim \rho_E} \log D(s, a) + \mathbb{E}_{(s,a) \sim \rho_{\pi}} \log(1 - D(s, a)) - \lambda H(\pi).$$

Reward for policy:

$$r_D(s, a) = -\log(1 - D(s, a)) \quad \text{или} \quad r_D(s, a) = \log D(s, a) - \log(1 - D(s, a)).$$

Optimal discriminator:

$$D^*(s, a) = \frac{\rho_E(s, a)}{\rho_E(s, a) + \rho_{\pi}(s, a)}.$$

14. Bandits, UCB, Thompson

Bandit value:

$$Q(a) = \mathbb{E}[r \mid a], \quad V^* = \max_a Q(a).$$

Regret:

$$\mathcal{R}_T = TV^* - \sum_{k=0}^{T-1} Q(a_k) = \sum_a n_T(a) (V^* - Q(a)).$$

Sample average:

$$Q_k(a) = \frac{\sum_{i=0}^k r_i \mathbf{1}\{a_i = a\}}{\sum_{i=0}^k \mathbf{1}\{a_i = a\}}, \quad Q_{k+1}(a) = Q_k(a) + \frac{1}{n_{k+1}(a)} (r_k - Q_k(a)).$$

Nonstationary:

$$Q_{k+1}(a) = Q_k(a) + \alpha(r_k - Q_k(a)).$$

Hoeffding:

$$\mathbb{P}(\mu \geq \hat{\mu} + u) \leq e^{-2nu^2}.$$

UCB:

$$a_k = \arg \max_a \left[Q_k(a) + c \sqrt{\frac{\log k}{n_k(a)}} \right].$$

Lai-Robbins lower bound:

$$\mathbb{E} \mathcal{R}_T \geq \log T \sum_{a \neq a^*} \frac{V^* - Q(a)}{\text{KL}(p(\cdot \mid a) \parallel p(\cdot \mid a^*))}.$$

Bayes update:

$$p(\theta_a \mid r) \propto p(r \mid \theta_a) p(\theta_a).$$

Thompson Sampling:

$$\theta_a \sim p(\theta_a \mid \mathcal{H}_k), \quad a_k = \arg \max_a \mathbb{E}[r \mid \theta_a].$$

Bernoulli-Beta:

$$r \sim \text{Bernoulli}(\theta), \quad \theta \sim \text{Beta}(\alpha, \beta), \\ \theta \mid r \sim \text{Beta}(\alpha + r, \beta + 1 - r), \quad \mathbb{E}\theta = \frac{\alpha}{\alpha + \beta}.$$

15. MCTS

Planning objective:

$$(a_0, a_1, \dots)^* \in \arg \max_{a_0, a_1, \dots} \mathbb{E}[R(\tau) \mid s_0, a_0, a_1, \dots].$$

Tree statistics:

$$N(s), \quad N(s, a), \quad Q(s, a).$$

UCT / tree policy:

$$a = \arg \max_a \left[Q(s, a) + C \sqrt{\frac{\log N(s)}{N(s, a)}} \right].$$

MCTS cycle:

$$\text{Selection} \rightarrow \text{Expansion} \rightarrow \text{Simulation} \rightarrow \text{Backpropagation}.$$

MC backup:

$$Q(s, a) \leftarrow Q(s, a) + \frac{1}{N(s, a)} (G - Q(s, a)), \quad N(s, a) \leftarrow N(s, a) + 1.$$

Policy from visits:

$$\pi(a \mid s) \propto N(s, a)^{1/T}.$$

AlphaZero-style PUCT:

$$a = \arg \max_a \left[Q(s, a) + c_{puct} P(s, a) \frac{\sqrt{N(s)}}{1 + N(s, a)} \right].$$

Value backup with neural value:

$$G \approx V_{\theta}(s_{leaf}) \quad \text{instead of random rollout.}$$

16. LQR, iLQR, Model-based RL

Model-based loop:

$$\mathcal{D} \leftarrow \mathcal{D} \cup \{(s, a, r, s')\}, \quad \hat{p}_{\theta}(s', r \mid s, a) \text{ train}, \quad \pi \leftarrow \text{planner}(\hat{p}_{\theta}, r).$$

Finite-horizon deterministic control:

$$\max_{a_0: T-1} \sum_{t=0}^{T-1} r(s_t, a_t), \quad s_{t+1} = f(s_t, a_t).$$

Direct differentiation:

$$\nabla_{a_t} J = \frac{\partial r_t}{\partial a_t} + \sum_{k>t} \frac{\partial r_k}{\partial s_k} \frac{\partial s_k}{\partial a_t}.$$

LQR dynamics/cost:

$$x_{t+1} = Ax_t + Bu_t,$$

$$J = \sum_{t=0}^{T-1} (x_t^{\top} Q x_t + u_t^{\top} R u_t) + x_T^{\top} Q_f x_T.$$

Riccati recursion:

$$P_T = Q_f,$$

$$K_t = (R + B^{\top} P_{t+1} B)^{-1} B^{\top} P_{t+1} A,$$

$$P_t = Q + A^{\top} P_{t+1} A - A^{\top} P_{t+1} B (R + B^{\top} P_{t+1} B)^{-1} B^{\top} P_{t+1} A.$$

Optimal control:

$$u_t^* = -K_t x_t.$$

iLQR local approximations:

$$x_{t+1} \approx f_t + F_x \delta x_t + F_u \delta u_t,$$

$$\ell(x, u) \approx \ell_0 + \ell_x^{\top} \delta x + \ell_u^{\top} \delta u + \frac{1}{2} \delta x^{\top} \ell_{xx} \delta x + \frac{1}{2} \delta u^{\top} \ell_{uu} \delta u + \delta u^{\top} \ell_{ux} \delta x.$$

Backward pass:

$$Q_u = \ell_u + F_u^{\top} V'_x, \quad Q_{uu} = \ell_{uu} + F_u^{\top} V'_{xx} F_u, \quad Q_{ux} = \ell_{ux} + F_u^{\top} V'_{xx} F_x,$$

$$k_t = -Q_{uu}^{-1} Q_u, \quad K_t = -Q_{uu}^{-1} Q_{ux}.$$

Forward pass:

$$u_t^{new} = u_t + \alpha k_t + K_t (x_t^{new} - x_t).$$

Stochastic model training:

$$\max_{\theta} \sum_{(s,a,s') \in \mathcal{D}} \log p_{\theta}(s' \mid s, a).$$